

Create Effective and Responsible AI User Experiences with The Human-AI Experience (HAX) Toolkit

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ABSTRACT

The HAX Toolkit (<https://aka.ms/haxtoolkit>) is a set of collaborative tools that helps teams working on user-facing AI plan, create, and evaluate human-AI user experiences. This course will help AI practitioners, human-AI interaction researchers, teachers, and students learn how to use the HAX Toolkit themselves and how to introduce it to others. The Toolkit is grounded in a set of Guidelines for Human-AI Interaction [1] that prescribe how AI systems should behave when interacting with people. Course attendees will explore the nuances of each guideline and learn how to use the AI patterns and examples in the HAX Design Library to apply the Guidelines. Course attendees will also learn how to guide cross-disciplinary teams in planning user-facing AI systems by using the HAX Workbook. For NLP systems, course attendees will learn to use the HAX Playbook [2] to anticipate and design for failures.

The HAX Toolkit is a set of collaborative tools that helps teams working on user-facing AI plan, create, and evaluate human-AI user experiences. This course will help AI practitioners, human-AI interaction researchers, teachers, and students learn how to use the HAX Toolkit themselves and how to introduce it to others. Course attendees will learn the nuances of the Guidelines for Human-AI Interaction, how to lead cross-disciplinary teams in planning human-AI interaction using the HAX Workbook, and how to use the HAX Workbook to plan for failures of NLP systems.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Interaction design process and methods;

KEYWORDS

artificial intelligence, human-AI interaction, responsible AI

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1 BENEFITS

The learning outcomes including skills and knowledge the attendees will gain as a result of attending this Course. This should also

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include the reasons that CHI attendees would want to take your course.

As a result of taking this course, attendees will be able to take a systematic, research-informed approach to creating effective and responsible experiences with AI. Specifically, attendees will be able to:

- Explain the nuances of the Guidelines for Human-AI Interaction
- Apply the Guidelines for Human-AI interaction for planning and evaluating user-facing AI systems
- Lead cross-disciplinary teams through planning user-facing AI systems with the HAX Workbook
- Plan for failures in NLP systems using the HAX Playbook

2 INTENDED AUDIENCE(S)

AI practitioners in various disciplines (project management, data science, UX) will benefit from this course by being able to immediately apply the learnings to creating user experiences with AI and onboarding their teams and colleagues to the HAX Toolkit.

Students interested in human-AI interaction will benefit from this course by being able to use the HAX Toolkit for class projects and in their future careers.

Professors who teach human-AI interaction will benefit from this course by being able to integrate the HAX Toolkit in their courses.

Researchers will benefit from this course by being able to build upon the HAX Toolkit and advance the UX of AI discipline.

3 PREREQUISITES

Basic familiarity with AI systems – what they are, what they can do. No data science background required.

4 CONTENT

4.1 Introduction

- Brief explanation of user-facing AI
- Discussion of how designing human-AI interaction is different from interaction with traditional software and why we need guidance specific to AI
- Overview of this course's learning outcomes

4.2 The Guidelines for Human-AI Interaction

- Introduction to each of the Guidelines
- Hands-on exercise: Find examples of the Guidelines
- Discussion of identified examples to help clarify the nuances of the Guidelines
- Discussion of when in the AI development lifecycle to use the Guidelines

4.3 The HAX Workbook

- Introduction to the purpose of the HAX Workbook: A planning and discussion guide for selecting which Guidelines for Human-AI Interaction to apply, and how to apply them
- Who should use the HAX workbook: AI practitioners from different disciplines (project management, data science, UX, engineering) should go through it together
- How to introduce the HAX Workbook to your team
- The role of the moderator
- Hands-on exercise: Fill out the HAX Workbook for a fictional AI feature
- Discussion & debrief on filling out the HAX Workbook

4.4 Planning for NLP failures with the HAX Playbook

- Introduction to the purpose of the HAX Playbook: To anticipate how an envisioned NLP system might fail, so you can design and evaluate solutions for those failures
- HAX Playbook demo

4.5 Closing

- Summary of tools in the HAX Toolkit
- Q&A, next steps, ideas for future work

5 PRACTICAL WORK

- Hands-on exercise: Find examples of each guideline
- Hands-on exercise: Fill out the HAX Workbook for a fictional AI system

6 INSTRUCTOR BACKGROUND

Dr. Mihaela Vorvoreanu co-lead the development of the HAX Toolkit and has delivered numerous workshops in industry and academic settings on the research behind the HAX Toolkit and how to use the HAX Toolkit. She currently leads the HAX Toolkit effort at Microsoft. Mihaela is Director for UX research and responsible education in Aether, Microsoft's research and advisory group on responsible AI. She has a background in UX and Communication Studies. Before joining Microsoft, Mihaela was a tenured faculty member at Purdue University where she founded the undergraduate UX program and led the graduate UX program in the Purdue Polytechnic.

6.1 Related publications:

Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19). Association for Computing Machinery, New York, NY, USA, Paper 3, 1–13. <https://doi.org/10.1145/3290605.3300233>

Tianyi Li, Mihaela Vorvoreanu, Derek DeBellis, and Saleema Amershi. 2022. Assessing Human-AI Interaction Early through Factorial Surveys: A Study on the Guidelines for Human-AI Interaction. ACM Transactions on Computer-Human Interaction. <https://doi.org/10.1145/3511605>

7 RESOURCES

HAX Toolkit: <https://aka.ms/haxtoolkit>

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8 ACCESSIBILITY

Course participants will work with the HAX Toolkit website, which has undergone Microsoft accessibility approvals as part of its release. The course will be online-only, which better supports the use of accessibility-supporting technologies such as live transcription.

ACKNOWLEDGMENTS

The course instructor would like to acknowledge the many co-authors and colleagues who have contributed to the research behind the HAX Toolkit and the HAX Toolkit website.

REFERENCES

- [1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19). Association for Computing Machinery, New York, NY, USA, Paper 3, 1–13. <https://doi.org/10.1145/3290605.3300233>
- [2] Matthew K. Hong, Adam Fourney, Derek DeBellis, and Saleema Amershi. 2021. Planning for Natural Language Failures with the AI Playbook. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 386, 1–11. <https://doi.org/10.1145/3411764.3445735>